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Electronic device comprising rollable display

Abstract

An electronic device is provided. The electronic device includes a rollable display, includes a rotation shaft arranged in an inner space of a housing in a first direction, a first printed circuit board (PCB) coupled to the rotation shaft, a rollable display arranged to be wound in the inner space of the housing, and is drawn out from the inside to the outside of the housing in conjunction with the rotation of the rotation shaft and in a second direction perpendicular to the first direction, a first electrical component which is arranged in the inner space of the housing and between a first flat portion and one end of the rotation shaft, fixed irrespective of the rotation of the rotation shaft, and a first rollable flexible PCB (FPCB) which allows the first PCB and the first electrical component to be electrically connected to each other.

Inventors: **Jung; Hojin (Suwon-si, KR), Ji; Youngmin (Suwon-si, KR), Kim; Junhyuk (Suwon-si, KR), Lee; Wonho (Suwon-si, KR), Kim; Yangwook (Suwon-si, KR), Cho; Hyoungtak (Suwon-si, KR), Hong; Hyunju (Suwon-si, KR)**

Applicant: **Samsung Electronics Co., Ltd.** (Suwon-si, KR)

Family ID: **81709034**

Assignee: **Samsung Electronics Co., Ltd. (Suwon-si, KR)**

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Primary Examiner: Parker; Allen L

Assistant Examiner: Kim; Keon

Attorney, Agent or Firm: Jefferson IP Law, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application is a continuation application, claiming priority under § 365 (c), of an International application No. PCT/KR2021/014258, filed on Oct. 14, 2021, which is based on and claims the benefit of a Korean patent application number 10-2020-0154381, filed on Nov. 18, 2020, in the Korean Intellectual Property Office, and of a Korean patent application number 10-2021-0048725, filed on Apr. 14, 2021, in the Korean Intellectual Property Office, the disclosure of each of which is incorporated by reference herein in its entirety.

BACKGROUND
1. Field
(1) The disclosure relates to an electronic device including a rollable display.
2. Description of Related Art
(2) With the development of display technology, research and development of electronic devices having a rollable display (or a flexible display) are being actively conducted.
(3) Electronic devices are being gradually transformed from a uniform rectangular shape into various shapes. For example, electronic devices are being researched and developed to have a form factor in which a display is capable of being folded, bent, rolled, or unfolded by applying a rollable display to the electronic devices.

(4) The above information is presented as background information only to assist with an understanding of the disclosure. No determination has been made, and no assertion is made, as to whether any of the above might be applicable as prior art with regard to the disclosure.

SUMMARY

(5) An electronic device may be designed to include, as a novel form factor, a hollow housing such that a rollable display is rolled inside the hollow housing. Such an electronic device may be designed such that the rollable display is drawn out from the inside to the outside of the hollow housing based on a predetermined event. In an electronic device including a rollable display, since at least a portion of the radiation area of an antenna is covered by the display, there may be difficulties in securing the performance of the antenna. Since most of the housing is covered by the display, a space for placing an antenna for cellular communication or short-range communication may be insufficient.

(6) Aspects of the disclosure are to address at least the above-mentioned problems and/or disadvantages and to provide at least the advantages described below. Accordingly, an aspect of the disclosure is to provide an electronic device including a hollow housing and a rollable display configured to be pushed out from inside the hollow housing, wherein an antenna radiator for cellular communication or short-range communication (e.g., Wi-Fi) is configured by using at least a portion of the housing.

(7) Additional aspects will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.

(8) In accordance with an aspect of the disclosure, an electronic device is provided. The electronic device includes a housing including a first flat portion disposed at one end of the housing and a second flat portion disposed at the other end of the housing in parallel to the first flat portion, wherein each of the first flat portion and the second flat portion includes at least one conductive portion and at least one non-conductive portion, a rotary shaft disposed in a first direction in an inner space of the housing, a first printed circuit board (PCB) coupled to the rotary shaft and configured to rotate following rotation of the rotary shaft, a rollable display arranged to be rolled in the inner space of the housing and pushed out from inside to outside of the housing in interlocking with the rotation of the rotary shaft in a second direction perpendicular to the first direction, a first electrical component disposed between the first flat portion and one end of the rotary shaft in the inner space of the housing and fixed regardless of the rotation of the rotary shaft, wherein the first electrical component is operably connected to the at least one conductive portion of the first flat portion via a first connecting member, and a first rollable flexible printed circuit board (FPCB) electrically connecting the first PCB and the first electrical component to each other, wherein the first rollable FPCB is spirally rolled around the one end of the rotary shaft.

(9) Various embodiments of the disclosure provide an electronic device having a novel form factor and including a hollow housing and a rollable display designed to be pushed out from the inside of the housing, wherein an antenna radiator for cellular communication or short-range communication (e.g., Wi-Fi) is configured by using at least a portion of the housing. As a result, it is possible to secure a space for arranging antennas.

(10) Other aspects, advantages, and salient features of the disclosure will become apparent to those skilled in the art from the following detailed description, which, taken in conjunction with the annexed drawings, discloses various embodiments of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other aspects, features, and advantages of certain embodiments of the disclosure will be more apparent from the following description taken in conjunction with the accompanying

drawings, in which:
(2) FIG. **1** is a block diagram of an electronic device in a network environment according to an embodiment of the disclosure;
(3) FIG. **2** is a front view illustrating an electronic device in a first state (e.g., a slide-in state) according to an embodiment of the disclosure;
(4) FIG. **3** is a front view illustrating an electronic device in the second state (e.g., a slide-out state) according to an embodiment of the disclosure;
(5) FIG. **4** is a rear view illustrating an electronic device in a first state according to an embodiment of the disclosure;
(6) FIG. **5** is a rear view illustrating an electronic device in a second state according to an embodiment of the disclosure;
(7) FIG. **6** is a front perspective view illustrating an electronic device in a first state according to an embodiment of the disclosure;
(8) FIG. **7** is a front perspective view illustrating an electronic device in a second state according to an embodiment of the disclosure;
(9) FIG. **8**A is an exploded perspective view of an electronic device according to an embodiment of the disclosure;
(10) FIG. **8**B is a perspective view illustrating a state in which a printed circuit board and a battery are coupled to each other according to an embodiment of the disclosure;
(11) FIG. **9** is a front perspective view illustrating a contact structure of an antenna of an electronic device according to an embodiment of the disclosure;
(12) FIG. **10** is a rear perspective view illustrating a contact structure of an antenna of an electronic device according to an embodiment of the disclosure;
(13) FIG. **11**A is a perspective view of an electronic device illustrating guide wheels configured to guide a movement of a rollable FPCB according to an embodiment of the disclosure;
(14) FIG. **11**B is a cross-sectional view of an electronic device illustrating a cross section of a guide wheel according to an embodiment of the disclosure;
(15) FIG. **12** is a plan view schematically illustrating a structure of a rollable FPCB according to an embodiment of the disclosure;
(16) FIG. **13** is a view illustrating a structure of an antenna of an electronic device according to an embodiment of the disclosure;
(17) FIG. **14** is a view illustrating a structure of an antenna of an electronic device according to an embodiment of the disclosure;
(18) FIG. **15** is a view illustrating a structure of an antenna of an electronic device according to an embodiment of the disclosure; and
(19) FIG. **16** is a plan view illustrating physical buttons of an electronic device according to an embodiment of the disclosure.
(20) The same reference numerals are used to represent the same elements throughout the drawings.
DETAILED DESCRIPTION
(21) The following description with reference to the accompanying drawings is provided to assist in a comprehensive understanding of various embodiments of the disclosure as defined by the claims and their equivalents. It includes various specific details to assist in that understanding but these are to be regarded as merely exemplary. Accordingly, those of ordinary skill in the art will recognize that various changes and modifications of the various embodiments described herein can be made without departing from the scope and spirit of the disclosure. In addition, descriptions of well-known functions and constructions may be omitted for clarity and conciseness.
(22) The terms and words used in the following description and claims are not limited to the bibliographical meanings, but, are merely used by the inventor to enable a clear and consistent understanding of the disclosure. Accordingly, it should be apparent to those skilled in the art that

the following description of various embodiments of the disclosure is provided for illustration purpose only and not for the purpose of limiting the disclosure as defined by the appended claims and their equivalents.

(23) It is to be understood that the singular forms "a," "an," and "the" include plural referents unless the context clearly dictates otherwise. Thus, for example, reference to "a component surface" includes reference to one or more of such surfaces.

(24) FIG. **1** is a block diagram illustrating an electronic device in a network environment according to an embodiment of the disclosure.

(25) Referring to FIG. **1**, an electronic device **101** in a network environment **100** may communicate with an external electronic device **102** via a first network **198** (e.g., a short-range wireless communication network), or at least one of an external electronic device **104** or a server **108** via a second network **199** (e.g., a long-range wireless communication network). According to an embodiment of the disclosure, the electronic device **101** may communicate with the external electronic device **104** via the server **108**. According to an embodiment of the disclosure, the electronic device **101** may include a processor **120**, a memory **130**, an input module **150**, a sound output module **155**, a display module **160**, an audio module **170**, a sensor module **176**, an interface **177**, a connecting terminal **178**, a haptic module **179**, a camera module **180**, a power management module **188**, a battery **189**, a communication module **190**, a subscriber identification module (SIM) **196**, or an antenna module **197**. In some embodiments of the disclosure, at least one of the components (e.g., the connecting terminal **178**) may be omitted from the electronic device **101**, or one or more other components may be added in the electronic device **101**. In some embodiments of the disclosure, some of the components (e.g., the sensor module **176**, the camera module **180**, or the antenna module **197**) may be implemented as a single component (e.g., the display module **160**).

(26) The processor **120** may execute, for example, software (e.g., a program **140**) to control at least one other component (e.g., a hardware or software component) of the electronic device **101** coupled with the processor **120**, and may perform various data processing or computation. According to one embodiment of the disclosure, as at least part of the data processing or computation, the processor **120** may store a command or data received from another component (e.g., the sensor module **176** or the communication module **190**) in a volatile memory **132**, process the command or the data stored in the volatile memory **132**, and store resulting data in a non-volatile memory **134**. According to an embodiment of the disclosure, the processor **120** may include a main processor **121** (e.g., a central processing unit (CPU) or an application processor (AP)), or an auxiliary processor **123** (e.g., a graphics processing unit (GPU), a neural processing unit (NPU), an image signal processor (ISP), a sensor hub processor, or a communication processor (CP)) that is operable independently from, or in conjunction with, the main processor **121**. For example, when the electronic device **101** includes the main processor **121** and the auxiliary processor **123**, the auxiliary processor **123** may be adapted to consume less power than the main processor **121**, or to be specific to a specified function. The auxiliary processor **123** may be implemented as separate from, or as part of the main processor **121**.

(27) The auxiliary processor **123** may control at least some of functions or states related to at least one component (e.g., the display module **160**, the sensor module **176**, or the communication module **190**) among the components of the electronic device **101**, instead of the main processor **121** while the main processor **121** is in an inactive (e.g., sleep) state, or together with the main processor **121** while the main processor **121** is in an active state (e.g., executing an application). According to an embodiment of the disclosure, the auxiliary processor **123** (e.g., an image signal processor or a communication processor) may be implemented as part of another component (e.g., the camera module **180** or the communication module **190**) functionally related to the auxiliary processor **123**. According to an embodiment of the disclosure, the auxiliary processor **123** (e.g., the neural processing unit) may include a hardware structure specified for artificial intelligence model

processing. An artificial intelligence model may be generated by machine learning. Such learning may be performed, e.g., by the electronic device **101** where the artificial intelligence is performed or via a separate server (e.g., the server **108**). Learning algorithms may include, but are not limited to, e.g., supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. The artificial intelligence model may include a plurality of artificial neural network layers. The artificial neural network may be a deep neural network (DNN), a convolutional neural network (CNN), a recurrent neural network (RNN), a restricted boltzmann machine (RBM), a deep belief network (DBN), a bidirectional recurrent deep neural network (BRDNN), deep Q-network or a combination of two or more thereof but is not limited thereto. The artificial intelligence model may, additionally or alternatively, include a software structure other than the hardware structure.
(28) The memory **130** may store various data used by at least one component (e.g., the processor **120** or the sensor module **176**) of the electronic device **101**. The various data may include, for example, software (e.g., the program **140**) and input data or output data for a command related thereto. The memory **130** may include the volatile memory **132** or the non-volatile memory **134**.
(29) The program **140** may be stored in the memory **130** as software, and may include, for example, an operating system (OS) **142**, middleware **144**, or an application **146**.
(30) The input module **150** may receive a command or data to be used by another component (e.g., the processor **120**) of the electronic device **101**, from the outside (e.g., a user) of the electronic device **101**. The input module **150** may include, for example, a microphone, a mouse, a keyboard, a key (e.g., a button), or a digital pen (e.g., a stylus pen).
(31) The sound output module **155** may output sound signals to the outside of the electronic device **101**. The sound output module **155** may include, for example, a speaker or a receiver. The speaker may be used for general purposes, such as playing multimedia or playing record. The receiver may be used for receiving incoming calls. According to an embodiment of the disclosure, the receiver may be implemented as separate from, or as part of the speaker.
(32) The display module **160** may visually provide information to the outside (e.g., a user) of the electronic device **101**. The display module **160** may include, for example, a display, a hologram device, or a projector and control circuitry to control a corresponding one of the display, hologram device, and projector. According to an embodiment of the disclosure, the display module **160** may include a touch sensor adapted to detect a touch, or a pressure sensor adapted to measure the intensity of force incurred by the touch.
(33) The audio module **170** may convert a sound into an electrical signal and vice versa. According to an embodiment of the disclosure, the audio module **170** may obtain the sound via the input module **150**, or output the sound via the sound output module **155** or a headphone of an external electronic device (e.g., the external electronic device **102**) directly (e.g., wiredly) or wirelessly coupled with the electronic device **101**.
(34) The sensor module **176** may detect an operational state (e.g., power or temperature) of the electronic device **101** or an environmental state (e.g., a state of a user) external to the electronic device **101**, and then generate an electrical signal or data value corresponding to the detected state. According to an embodiment of the disclosure, the sensor module **176** may include, for example, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a proximity sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an illuminance sensor.
(35) The interface **177** may support one or more specified protocols to be used for the electronic device **101** to be coupled with the external electronic device (e.g., the external electronic device **102**) directly (e.g., wiredly) or wirelessly. According to an embodiment of the disclosure, the interface **177** may include, for example, a high definition multimedia interface (HDMI), a universal serial bus (USB) interface, a secure digital (SD) card interface, or an audio interface.
(36) A connecting terminal **178** may include a connector via which the electronic device **101** may be physically connected with the external electronic device (e.g., the external electronic device

102). According to an embodiment of the disclosure, the connecting terminal **178** may include, for example, a HDMI connector, a USB connector, a SD card connector, or an audio connector (e.g., a headphone connector).

(37) The haptic module **179** may convert an electrical signal into a mechanical stimulus (e.g., a vibration or a movement) or electrical stimulus which may be recognized by a user via his tactile sensation or kinesthetic sensation. According to an embodiment of the disclosure, the haptic module **179** may include, for example, a motor, a piezoelectric element, or an electric stimulator.

(38) The camera module **180** may capture a still image or moving images. According to an embodiment of the disclosure, the camera module **180** may include one or more lenses, image sensors, image signal processors, or flashes.

(39) The power management module **188** may manage power supplied to the electronic device **101**. According to one embodiment of the disclosure, the power management module **188** may be implemented as at least part of, for example, a power management integrated circuit (PMIC).

(40) The battery **189** may supply power to at least one component of the electronic device **101**. According to an embodiment of the disclosure, the battery **189** may include, for example, a primary cell which is not rechargeable, a secondary cell which is rechargeable, or a fuel cell.

(41) The communication module **190** may support establishing a direct (e.g., wired) communication channel or a wireless communication channel between the electronic device **101** and the external electronic device (e.g., the external electronic device **102**, the external electronic device **104**, or the server **108**) and performing communication via the established communication channel. The communication module **190** may include one or more communication processors that are operable independently from the processor **120** (e.g., the application processor (AP)) and supports a direct (e.g., wired) communication or a wireless communication. According to an embodiment of the disclosure, the communication module **190** may include a wireless communication module **192** (e.g., a cellular communication module, a short-range wireless communication module, or a global navigation satellite system (GNSS) communication module) or a wired communication module **194** (e.g., a local area network (LAN) communication module or a power line communication (PLC) module). A corresponding one of these communication modules may communicate with the external electronic device via the first network **198** (e.g., a short-range communication network, such as Bluetooth™, wireless-fidelity (Wi-Fi) direct, or infrared data association (IrDA)) or the second network **199** (e.g., a long-range communication network, such as a legacy cellular network, a fifth-generation (5G) network, a next-generation communication network, the Internet, or a computer network (e.g., LAN or wide area network (WAN)). These various types of communication modules may be implemented as a single component (e.g., a single chip), or may be implemented as multi components (e.g., multi chips) separate from each other. The wireless communication module **192** may identify and authenticate the electronic device **101** in a communication network, such as the first network **198** or the second network **199**, using subscriber information (e.g., international mobile subscriber identity (IMSI)) stored in the SIM **196**.

(42) The wireless communication module **192** may support a 5G network, after a fourth-generation (4G) network, and next-generation communication technology, e.g., new radio (NR) access technology. The NR access technology may support enhanced mobile broadband (eMBB), massive machine type communications (mMTC), or ultra-reliable and low-latency communications (URLLC). The wireless communication module **192** may support a high-frequency band (e.g., the millimeter (mm) Wave band) to achieve, e.g., a high data transmission rate. The wireless communication module **192** may support various technologies for securing performance on a high-frequency band, such as, e.g., beamforming, massive multiple-input and multiple-output (massive MIMO), full dimensional MIMO (FD-MIMO), array antenna, analog beam-forming, or large scale antenna. The wireless communication module **192** may support various requirements specified in the electronic device **101**, an external electronic device (e.g., the external electronic device **104**), or

a network system (e.g., the second network **199**). According to an embodiment of the disclosure, the wireless communication module **192** may support a peak data rate (e.g., 20 Gbps or more) for implementing eMBB, loss coverage (e.g., 164 dB or less) for implementing mMTC, or U-plane latency (e.g., 0.5 ms or less for each of downlink (DL) and uplink (UL), or a round trip of 1 ms or less) for implementing URLLC.

(43) The antenna module **197** may transmit or receive a signal or power to or from the outside (e.g., the external electronic device) of the electronic device **101**. According to an embodiment of the disclosure, the antenna module **197** may include an antenna including a radiating element including a conductive material or a conductive pattern formed in or on a substrate (e.g., a printed circuit board (PCB)). According to an embodiment of the disclosure, the antenna module **197** may include a plurality of antennas (e.g., array antennas). In such a case, at least one antenna appropriate for a communication scheme used in the communication network, such as the first network **198** or the second network **199**, may be selected, for example, by the communication module **190** (e.g., the wireless communication module **192**) from the plurality of antennas. The signal or the power may then be transmitted or received between the communication module **190** and the external electronic device via the selected at least one antenna. According to an embodiment of the disclosure, another component (e.g., a radio frequency integrated circuit (RFIC)) other than the radiating element may be additionally formed as part of the antenna module **197**.

(44) According to various embodiments of the disclosure, the antenna module **197** may form a mmWave antenna module. According to an embodiment of the disclosure, the mmWave antenna module may include a printed circuit board, a RFIC disposed on a first surface (e.g., the bottom surface) of the printed circuit board, or adjacent to the first surface and capable of supporting a designated high-frequency band (e.g., the mmWave band), and a plurality of antennas (e.g., array antennas) disposed on a second surface (e.g., the top or a side surface) of the printed circuit board, or adjacent to the second surface and capable of transmitting or receiving signals of the designated high-frequency band.

(45) At least some of the above-described components may be coupled mutually and communicate signals (e.g., commands or data) therebetween via an inter-peripheral communication scheme (e.g., a bus, general purpose input and output (GPIO), serial peripheral interface (SPI), or mobile industry processor interface (MIPI)).

(46) According to an embodiment of the disclosure, commands or data may be transmitted or received between the electronic device **101** and the external electronic device **104** via the server **108** coupled with the second network **199**. Each of the external electronic devices **102** or **104** may be a device of a same type as, or a different type, from the electronic device **101**. According to an embodiment of the disclosure, all or some of operations to be executed at the electronic device **101** may be executed at one or more of the external electronic devices **102**, **104**, or **108**. For example, if the electronic device **101** should perform a function or a service automatically, or in response to a request from a user or another device, the electronic device **101**, instead of, or in addition to, executing the function or the service, may request the one or more external electronic devices to perform at least part of the function or the service. The one or more external electronic devices receiving the request may perform the at least part of the function or the service requested, or an additional function or an additional service related to the request, and transfer an outcome of the performing to the electronic device **101**. The electronic device **101** may provide the outcome, with or without further processing of the outcome, as at least part of a reply to the request. To that end, a cloud computing, distributed computing, mobile edge computing (MEC), or client-server computing technology may be used, for example. The electronic device **101** may provide ultra low-latency services using, e.g., distributed computing or mobile edge computing. In another embodiment of the disclosure, the external electronic device **104** may include an internet-of-things (IoT) device. The server **108** may be an intelligent server using machine learning and/or a neural network. According to an embodiment of the disclosure, the external electronic device **104** or the

server **108** may be included in the second network **199**. The electronic device **101** may be applied to intelligent services (e.g., smart home, smart city, smart car, or healthcare) based on 5G communication technology or IoT-related technology.

(47) The electronic device according to various embodiments of the disclosure may be one of various types of electronic devices. The electronic devices may include, for example, a portable communication device (e.g., a smartphone), a computer device, a portable multimedia device, a portable medical device, a camera, a wearable device, or a home appliance. The electronic devices according to embodiments of the disclosure are not limited to those described above.

(48) It should be appreciated that various embodiments of the disclosure and the terms used therein are not intended to limit the technological features set forth herein to particular embodiments and include various changes, equivalents, or replacements for a corresponding embodiment. With regard to the description of the drawings, similar reference numerals may be used to refer to similar or related elements. It is to be understood that a singular form of a noun corresponding to an item may include one or more of the things, unless the relevant context clearly indicates otherwise. As used herein, each of such phrases as “A or B”, “at least one of A and B”, “at least one of A or B”, “A, B, or C”, “at least one of A, B, and C”, and “at least one of A, B, or C” may include any one of, or all possible combinations of the items enumerated together in a corresponding one of the phrases. As used herein, such terms as “1st” and “2nd”, or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (e.g., importance or order). It is to be understood that if an element (e.g., a first element) is referred to, with or without the term “operatively” or “communicatively”, as “coupled with”, “coupled to”, “connected with”, or “connected to” another element (e.g., a second element), it means that the element may be coupled with the other element directly (e.g., wiredly), wirelessly, or via a third element.

(49) As used in connection with various embodiments of the disclosure, the term “module” may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, logic, logic block, part, or circuitry. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment of the disclosure, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

(50) Various embodiments as set forth herein may be implemented as software (e.g., the program **140**) including one or more instructions that are stored in a storage medium (e.g., an internal memory **136** or an external memory **138**) that is readable by a machine (e.g., the electronic device **101**). For example, a processor (e.g., the processor **120**) of the machine (e.g., the electronic device **101**) may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term “non-transitory” simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

(51) According to an embodiment of the disclosure, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., a compact disc read only memory (CD-ROM)), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at

least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

(52) According to various embodiments of the disclosure, each component (e.g., module or program) of the above-described components may include a singular or a plurality of entities, and some of the plurality of entities may be separately disposed in any other component. According to various embodiments of the disclosure, one or more components or operations among the above-described components may be omitted, or one or more other components or operations may be added. Alternatively or additionally, a plurality of components (e.g., module or program) may be integrated into one component. In this case, the integrated component may perform one or more functions of each component of the plurality of components identically or similarly to those performed by the corresponding component among the plurality of components prior to the integration. According to various embodiments of the disclosure, operations performed by a module, program, or other component may be executed sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

(53) According to various embodiments of the disclosure, an electronic device (e.g., the electronic device **101** of FIG. **1**) may include a housing (e.g., a housing **200** in FIG. **2**) including a first flat portion (e.g., a first flat portion **220** in FIG. **2**) disposed at one end of the housing **200**, and a second flat portion (e.g., a second flat portion **230** in FIG. **2**) disposed at the other end of the housing **200** and disposed in parallel to the first flat portion **220**, wherein each of the first flat portion **220** and the second flat portion **230** includes at least one conductive portion (e.g., a first conductive portion **611**, a second conductive portion **612**, and/or the third conductive portion **613** in FIG. **6**) and at least one non-conductive portion (e.g., a non-conductive portion **620** in FIG. **6**), a rotary shaft **811** disposed in a first direction in the inner space **801** of the housing **200**, a first PCB **821** coupled to the rotary shaft **811** and configured to rotate following the rotation of the rotary shaft **811**, a rollable display **241**, also referred to as a first display **241**, arranged to be rolled in the inner space **801** of the housing **200** and pushed out from inside to outside of the housing **200** in interlocking with the rotation of the rotary shaft **811** in a second direction perpendicular to the first direction, a first electrical component **822** (e.g., a second PCB **822** in FIG. **8**A) disposed between the first flat portion **220** and one end of the rotary shaft **811** in the inner space **801** of the housing **200** and fixed regardless of the rotation of the rotary shaft **811**, wherein the first electrical component **822** is operably connected to the conductive portion (e.g., the first conductive portion **611**, the second conductive portion **612**, and/or the third conductive portion **613** in FIG. **6**) of the first flat portion **220** via a first connecting member **911**, and a first rollable FPCB **831** electrically connecting the first PCB **821** and the first electrical component to each other, wherein the first rollable FPCB **831** is spirally rolled around the one end of the rotary shaft **811**.

(54) According to an embodiment of the disclosure, one end of the first rollable FPCB **831** may be disposed adjacent to the one end of the rotary shaft **811** and may be electrically connected to a portion of the first PCB **821**, and the other end of the first rollable FPCB **831** may be electrically connected to a portion of the first electrical component **822**.

(55) According to an embodiment of the disclosure, the first electrical component **822** may include a circular first opening **1011** in the center thereof, and the first rollable FPCB **831** may have a diameter smaller than or equal to that of the first opening **1011**.

(56) According to an embodiment of the disclosure, a first shield member may be disposed between the first rollable FPCB **831** and the first PCB **821**, and the first rollable FPCB **831** may be configured to form a radiation pattern in a direction from the first shield member toward the first flat portion **220** by including at least one wire configured to provide a wireless charging function.

(57) According to an embodiment of the disclosure, the conductive portion (e.g., the first conductive portion **611**, the second conductive portion **612**, and/or the third conductive portion **613** in FIG. **6**) may include the first conductive portion **611** used as a first antenna configured to

perform wireless communication with a predetermined first frequency, the second conductive portion **612** used as a second antenna configured to perform wireless communication with a predetermined second frequency, and a third conductive portion **613** used as a third antenna configured to perform wireless communication with a predetermined third frequency.
(58) According to an embodiment of the disclosure, the rotary shaft **811** may be attached to one surface of the first PCB **821**.
(59) According to an embodiment of the disclosure, the housing **200** may be configured in a hollow shape which includes one of a cylindrical shape and a prismatic shape, and the rollable display **241** may be accommodated in the inner space **801** of the housing **200** in a rolled state.
(60) According to an embodiment of the disclosure, the electronic device may further include a sub-display **242** visually exposed outside through a portion of the housing **200**.
(61) According to an embodiment of the disclosure, the electronic device may further include a second electrical component **823** disposed between the second flat portion **230** and the other end of the rotary shaft **811** in the inner space **801** of the housing **200** and fixed regardless of the rotation of the rotary shaft **811**, wherein the second electrical component **823** is operably connected to the conductive portion of the second flat portion **230** via a second connecting member, and a second rollable FPCB **832** electrically connecting the first PCB **821** and the second electrical component **823** to each other, wherein the second rollable FPCB **832** is spirally rolled around the other end of the rotary shaft **811**.
(62) According to an embodiment of the disclosure, the first electric component **822** may be a second PCB, and the second electric component **823** may be a third PCB.
(63) According to an embodiment of the disclosure, one end of the second rollable FPCB **832** may be disposed adjacent to the other end of the rotary shaft **811** and may be electrically connected to a portion of the first PCB **821**, and the other end of the second rollable FPCB **832** may be electrically connected to a portion of the second electrical component **823**.
(64) According to an embodiment of the disclosure, the third PCB may include a circular second opening in a center thereof, and the second rollable FPCB **832** may have a diameter smaller than or equal to that of the second opening.
(65) According to an embodiment of the disclosure, a second shield member may be disposed between the second rollable FPCB **832** and the first PCB **821**.
(66) According to an embodiment of the disclosure, the second rollable FPCB **832** may be configured to form a radiation pattern in a direction from the second shield member toward the second flat portion **230** by including at least one wire configured to provide a wireless charging function.
(67) According to an embodiment of the disclosure, the electronic device may further include a plurality of guide wheels spirally disposed along a surface of the first rollable PCB at intervals, and each of the plurality of guide wheels may include a wheel member that comes into contact with the surface of the first rollable PCB and a guide shaft **811** penetrating the wheel member.
(68) According to an embodiment of the disclosure, the electronic device may further include a plurality of rail members configured to guide movement of the plurality of guide wheels, and the plurality of rail members are provided to be directed to the center of the rotary shaft **811** to enable the plurality of guide wheels to reciprocate toward the center of the rotary shaft **811**.
(69) According to an embodiment of the disclosure, the plurality of rail members (e.g., rail members **1120** in FIG. **11**A) may include at least one first rail member (e.g., a first rail member **1121** in FIG. **11**B) provided on the first flat portion **220**, and a second rail member (e.g., a second rail member **1122** in FIG. **11**B) provided on the first shield member disposed between the first rollable FPCB **831** and the first PCB **821**.
(70) According to an embodiment of the disclosure, the conductive portion may include the first conductive portion **611**, the second conductive portion **612**, and a third conductive portion **613** disposed at intervals along the outer edge of the first flat portion **220** to be split by the non-

conductive portion.

(71) According to an embodiment of the disclosure, the first conductive portion **611** may be configured to operate as a first antenna by including a first feeding point (e.g., a first feeding point **941-1** in FIG. **9**) electrically connected to a feeding portion of the first electrical component **822** and a first ground point (e.g., a first ground point **942-1** in FIG. **9**) electrically connected to a ground, the second conductive portion **612** may be configured to operate as a second antenna by including a second feeding point (e.g., a second feeding point **941-2** in FIG. **9**) electrically connected to the feeding portion of the first electrical component **822** and a second ground point (e.g., a second ground point **942-2**) electrically connected to the ground, and the third conductive portion **613** may be configured to operate as a third antenna by including a third feeding point (e.g., a third feeding point **941-3** in FIG. **9**) electrically connected to the feeding portion of the first electrical component **822** and a switch portion electrically connected to a switch of the first electrical component **822**.

(72) According to an embodiment of the disclosure, the first conductive portion **611** to the third conductive portion **613** may at least partially include extensions (e.g., the extensions **611***b*, **612***b*, and **613***b* of FIG. **14**) extending toward the rotary shaft **811**.

(73) FIG. **2** is a front view of an electronic device in a first state (e.g., a slide-in state) according to an embodiment of the disclosure.

(74) FIG. **3** is a front view of an electronic device in the second state (e.g., a slide-out state) according to an embodiment of the disclosure.

(75) FIG. **4** is a rear view of an electronic device in a first state according to an embodiment of the disclosure.

(76) FIG. **5** is a rear view of an electronic device in a second state according to an embodiment of the disclosure.

(77) Referring to FIGS. **2** to **5**, the electronic device according to various embodiments may include a hollow housing **200** according to an embodiment of the disclosure.

(78) According to an embodiment of the disclosure, the hollow housing **200** may include a main body **210** having a hollow (e.g., the space **801** in FIG. **8**A) and having a predetermined length, the first flat portion **220** disposed at one end **210***a* of the main body **210** to be coupled to the main body **210** or configured integrally with the main body **210**, and a second flat portion **230** disposed at the other end **210***b* of the main body **210** to be coupled to the main body **210** or configured integrally with the main body **210**. According to an embodiment of the disclosure, the main body **210** may have a cylindrical shape, but at least a portion of its surface may have a flat surface without curvature. In an embodiment of the disclosure, the first flat portion **220** may be provided based on the shape of the one end **210***a* of the main body **210**. For example, when the one end **210***a* of the main body **210** has a circular shape, the first flat portion **220** may have a circular shape corresponding to the one end **210***a*. As another example, the second flat portion **230** may be provided based on the shape of the other end **210***b* of the main body **210**. For example, when the other end **210***b* of the main body **210** has a circular shape, the second flat portion **230** may have a circular shape corresponding to the other end **210***b*.

(79) The hollow mentioned herein may include a cylindrical shape, a prismatic, a square pillar shape, a triangular prismatic shape, or the like.

(80) According to an embodiment of the disclosure, the first flat portion **220** may extend from the one end **210***a* of the main body **210**. According to an embodiment of the disclosure, the second flat portion may extend from the other end **210***b* of the main body **210**. According to an embodiment of the disclosure, the first flat portion **220** and the second flat portion **230** may be disposed to be substantially parallel to each other. According to an embodiment of the disclosure, the main body **210** may be disposed between the first flat portion **220** and the second flat portion **230**, thereby substantially defining side surface portions **211** of the electronic device **101**. For example, the main body **210** may define the side surface portions **211** of the electronic device **101** each of which

includes a curved surface. According to various embodiments of the disclosure, the curvature of the main body **210** or the diameter of the main body **210** may be variously designed and modified. Although not illustrated, the main body **210** may have a substantially cylindrical shape, but may at least partially include a flat plane.

(81) According to an embodiment of the disclosure, the electronic device **101** may include a second display **242** exposed outside through at least a portion of the main body **210** of the housing **200**. According to an embodiment of the disclosure, the second display **242** may be always visually exposed outside regardless of whether the electronic device **101** is in a first state or in a second state. For example, the second display **242** may be always visually exposed outside by being exposed outside through at least a portion of the surface of the main body **210**.

(82) According to an embodiment of the disclosure, the second display **242** may be disposed in a recess (not illustrated) provided on at least a portion of the surface of the main body **210** to occupy a portion of the area of the main body **210**. According to an embodiment of the disclosure, since the surface of the main body **210** has a curved surface, the second display **242** may include a curved surface having a similar curvature to that of the surface of the main body **210**. In some embodiments of the disclosure, the second display **242** may have a flat shape, in which case, a flat support surface (not illustrated) may be provided on the surface of the main body **210** to support the second display **242**.

(83) According to an embodiment of the disclosure, camera modules **180***a* and **180***b* (e.g., the camera module of FIG. **1**) may be disposed in the main body **210** of the housing **200**. According to an embodiment of the disclosure, the camera modules **180***a* and **180***b* may include a first camera module **180***a* arranged to face the front side of the electronic device **101** (e.g., the z direction) where the second display **242** is visually exposed and a second camera module **180***b* disposed to face the rear side of the electronic device **101** (e.g., the −z direction) arranged to a direction opposite to the front side. According to some embodiments of the disclosure, one of the first camera module **180***a* and the second camera module **180***b* may be omitted.

(84) According to an embodiment of the disclosure, the camera modules **180***a* and **180***b* may include one lens or two or more lenses (e.g., a wide-angle lens, an ultra-wide-angle lens, or a telephoto lens) and image sensors. In some embodiments of the disclosure, the camera modules may include a LiDAR sensor, a time-of-flight (TOF) lens, and/or image sensor.

(85) According to various embodiments of the disclosure, the first camera module **180***a* arranged to face the front side (e.g., the z-direction) of the electronic device **101** through which the second display **242** is visually exposed may be arranged in the inner space of the electronic device **101** to be in contact with the external environment through an opening perforated through the second display **242** or a transmission area. According to an embodiment of the disclosure, the area of the second display **242** that faces the first camera module **305** may be configured with a portion of a content display area as the transmission area having a predetermined transmittance. According to an embodiment of the disclosure, the transmission area may have a transmittance ranging from about 5% to about 20%. The transmission area may include an area overlapping an effective area (e.g., a field of view (FOV)) of the camera module through which light captured by an image sensor to generate an image passes. For example, the transmission area of the second display **242** may include an area having a lower pixel density and/or a lower wiring density than the surrounding area. For example, the first camera module **180***a* may include an under-display camera (UDC).

(86) According to an embodiment of the disclosure, the main body **210** of the housing **200** may include a slit-shaped hole **213** through which the first display **241** is pushed out from the inner space (e.g., the space **801** of FIG. **8**A) of the housing **200**. According to an embodiment of the disclosure, the slit-shaped hole **213** may extend in a first direction (e.g., the y direction), which is the length direction of the main body **210**, and may have a first length d**1**. One end of the first display **241** may be connected to a rotary shaft **811** (e.g., the rotary shaft **811** of FIGS. **8**A and **8**B)

disposed in the inner space **801** of the housing **200** and may be slid out from the inner space **801** of the housing **200** in interlocking with the rotation of the rotary shaft **811**.
(87) According to an embodiment of the disclosure, in the second state of the electronic device **101**, the first display **241** may be pushed out from the main body **210** of the housing **200** by a predetermined second length d**2** to a second direction (e.g., the x direction) perpendicular to the first direction (e.g., the y direction). For example, in the second state of the electronic device **101**, the area of the display area of the first display **241** may have a horizontal width corresponding to the second length d**2** and a vertical width corresponding to the first length d**1**.
(88) According to an embodiment of the disclosure, the portion of the first display **241** visually seen from outside in the second state of the electronic device **101** may be supported by a display support structure. According to an embodiment of the disclosure, the display support structure may include a bendable member or a bendable support member (not illustrated) (e.g., an articulated hinge module) and/or a handler member **250** coupled to an end of the first display **241** and extending in the first direction (e.g., the y direction).
(89) According to an embodiment of the disclosure, the bendable member may be a component including a plurality of hinge bars (not illustrated) arranged in the first direction (e.g., the y direction) perpendicular to the second direction (e.g., the x direction) where the first display **241** slides. A bendable member according to various embodiments may be referred to as an articulated hinge, a multi-bar assembly, or a hinge bar assembly. According to an embodiment of the disclosure, the bendable member may be at least partially drawn into or pushed out from the inner space **801** of the housing **200** while supporting the first display **241**. In some embodiments of the disclosure, the bendable member may be omitted.
(90) According to an embodiment of the disclosure, the handler member **250** may be coupled to an end of the bendable member and disposed in the first direction (e.g., the y direction) which is the length direction of the main body **210**. According to an embodiment of the disclosure, the handler member **250** may serve to prevent the first display **241** from being distorted to the z direction or the −z direction in the second state of the electronic device **101**. According to an embodiment of the disclosure, the handler member **250** may serve as a handle for a user to hold a portion of the first display **241**. According to an embodiment of the disclosure, the handler member **250** may include at least one sensor configured to detect a user's holding or a state (e.g., the posture or direction) of the electronic device **101**. According to an embodiment of the disclosure, the handler member **250** may further include an antenna or a substrate (not illustrated).
(91) According to an embodiment of the disclosure, in the first state of the electronic device **101**, the first display **241** may slide to a third direction (the −x direction) opposite to the second direction (the x direction) to be slid into the inner space **801** of the housing **200**.
(92) According to an embodiment of the disclosure, in the first state of the electronic device **101**, at least a portion of the first display **241** may be disposed in a state in which at least a portion of the first display **241** is rolled inside the main body **210** of the housing **200**. For example, when the electronic device **101** is switched from the second state to the first state, the first display **241** may be inserted into the inner space **801** of the housing **200** through the slit-shaped hole **213** while moving in the third direction (the −x direction) opposite to the second direction (the x direction). One end of the first display **241** is connected to the rotary shaft **811** disposed in the inner space **801** of the housing **200** and may be introduced into the inner space **801** of the housing **200** in interlocking with the rotation of the rotary shaft **811**. According to an embodiment of the disclosure, in the first state, at least a portion of the first display **241** disposed inside the housing **200** may be rolled in a jelly-roll shape.
(93) According to an embodiment of the disclosure, the housing **200** may provide the inner space **801** in which various components (e.g., a printed circuit board, an antenna module (e.g., the antenna module **197** of FIG. **1**), a sensor module (e.g., the sensor module **176** of FIG. **1**) or a battery (e.g., the battery **189** in FIG. **1**) of the electronic device **101** may be arranged. In an

embodiment of the disclosure, at least some of the various components of the electronic device **101** arranged in the inner space **801** of the housing **200** may be visually exposed outside through the surface of the main body **210** of the housing **200**. According to an embodiment of the disclosure, the components visually exposed outside the main body **210** may include camera modules **180***a* and **180***b* (e.g., the camera module **180** of FIG. **1**), a sound output module (not illustrated) (e.g., the sound output module **155** of FIG. **1**), and/or sensors. According to an embodiment of the disclosure, the sensors may include at least one of a receiver, a proximity sensor, an ultrasonic sensor, a gesture sensor, a gyro sensor, an atmospheric pressure sensor, a magnetic sensor, an acceleration sensor, a grip sensor, a color sensor, an infrared (IR) sensor, a biometric sensor, a temperature sensor, a humidity sensor, or an indicator. For example, at least some of the components may be disposed under the second display **242** or visually exposed outside through a partial area of the second display **242**.
(94) Referring to FIGS. **2** and **3**, in various embodiments of the disclosure, the front surface of the electronic device **101** may be defined as a surface on which the second display **242** (e.g., a sub-display) of the electronic device **101** is visually visible from outside. For example, the front surface of the electronic device **101** may be disposed in the z direction which the surface of the second display **242** faces. According to an embodiment of the disclosure, the second display **242** may be visually visible through the front surface of the electronic device **101**.
(95) Referring to FIGS. **4** and **5**, in various embodiments of the disclosure, the rear surface of the electronic device **101** may be defined as a surface facing a direction opposite to the front surface. For example, the rear surface of the electronic device **101** may face the −z direction opposite to the z direction. According to an embodiment of the disclosure, the second display **242** may be invisible through the rear surface of the electronic device **101**.
(96) Referring to FIGS. **2** and **4**, in various embodiments of the disclosure, the first state (e.g., the slide-in state) of the electronic device **101** may include a state in which at least a portion of the first display **241** (e.g., a main display) is invisible from outside by being rolled inside the hollow housing **200**. For example, the first state may refer to the state in which the externally visible area of the first display **241** is the smallest. In various embodiments of the disclosure, the first state may be referred to as a pulled-in state, a slide-in state, or a contracted state.
(97) Referring to FIGS. **3** and **5**, in various embodiments of the disclosure, the second state (e.g., the slide-out state) of the electronic device **101** may include a state in which the first display **241** (e.g., the main display) is visible from outside as the first display **241** is pushed out to the outside of the hollow housing **200**. For example, the second state may refer to the state in which the externally visible area of the first display **241** is the largest. In various embodiments of the disclosure, the second state may be referred to as a pushed-out state, a slide-out state, or an extended state.
(98) In an embodiment of the disclosure, the first state may be referred to as a first shape, and the second state may be referred to as a second shape. For example, the first shape may include a normal state, a retracted state, a contracted state, a closed state, or a slide-in state, and the second shape may include an extended state, an open state, or a slide-out state.
(99) According to an embodiment of the disclosure, the electronic device **101** may be in a third state which is a state between the first state and the second state. For example, the third state may be referred to as a third shape, and the third shape may include a free-stop state or an intermediate state. For example, the intermediate state may refer to a state in which an area between the case where the externally visible area of the first display **241** is the smallest and the case where the externally visible area is the largest is visible from outside.
(100) At least one of the components described with reference to FIGS. **2** to **5** may be omitted from the electronic device **101** according to various embodiments or other components may be additionally included in the electronic device **101**.
(101) FIG. **6** is a front perspective view illustrating an electronic device in a first state according to

an embodiment of the disclosure.
(102) FIG. **7** is a front perspective view illustrating an electronic device in a second state according to an embodiment of the disclosure.
(103) The electronic device **101** illustrated in FIGS. **6** and **7** may include an embodiment that is at least partially similar to or different from the electronic device **101** illustrated in FIGS. **1** to **5**. Hereinafter, features of the electronic device **101** that have not been described with reference to FIGS. **1** to **5** or are changed from FIGS. **1** to **5** will be mainly described with reference to FIGS. **6** and **7**.
(104) Referring to FIGS. **6** and **7**, the first flat portion **220** and/or a second flat portion **230** of the electronic device **101** according to an embodiment may include a conductive portion **610** and/or the non-conductive portion **620**.
(105) According to an embodiment of the disclosure, the non-conductive portion **620** disposed on the first flat portion **220** and/or the second flat portion **230** may be an injection-molded product.
(106) According to an embodiment of the disclosure, the conductive portion **610** disposed on the first flat portion **220** and/or the second flat portion **230** may include a metal material, and may be used as an antenna radiator for cellular communication or short-range communication (e.g., Wi-Fi).
(107) According to an embodiment of the disclosure, the conductive portion **610** may include the first conductive portion **611** used as a first antenna, the second conductive portion **612** used as a second antenna, and/or a third conductive portion **613** used as a third antenna.
(108) According to an embodiment of the disclosure, the first conductive portion **611**, the second conductive portion **612**, and/or the third conductive portion **613** may be split by the non-conductive portion **620**.
(109) According to an embodiment of the disclosure, the first antenna may be an antenna configured to perform wireless communication with a predetermined first frequency.
(110) According to an embodiment of the disclosure, the second antenna may be an antenna configured to perform wireless communication with a predetermined second frequency.
(111) According to an embodiment of the disclosure, the third antenna may be an antenna configured to perform wireless communication with a predetermined third frequency.
(112) According to an embodiment of the disclosure, the first frequency may be a communication frequency of Sub-6 (6 GHz or less) band for 5G communication. According to an embodiment of the disclosure, the first frequency may further include a communication frequency for the second generation (2G), third-generation (3G), 4G, or long term evolution (LTE) network.
(113) According to an embodiment of the disclosure, the second frequency may be a communication frequency of a super-high frequency (e.g., mmWave (e.g., the 28-gigahertz (28 GHZ) or 39-gigahertz (39 GHZ)) band for 5G communication.
(114) According to an embodiment of the disclosure, the third frequency may be a communication frequency for the purpose of Bluetooth or Wi-Fi communication in the 2.4-GHz or 5-GHz ISM band.
(115) FIG. **8**A is an exploded perspective view of an electronic device according to an embodiment of the disclosure.
(116) FIG. **8**B is a perspective view illustrating a state in which a printed circuit board and a battery are coupled to each other according to an embodiment of the disclosure.
(117) FIG. **9** is a front perspective view illustrating a contact structure of an antenna of an electronic device according to an embodiment of the disclosure.
(118) Referring to FIGS. **8**A, **8**B, and **9**, the electronic device **101** may include an embodiment that is at least partially similar to or different from the electronic device **101** illustrated in FIGS. **1** to **7**. Hereinafter, features of the electronic device **101** that have not been described with reference to FIGS. **1** to **7** or are changed from FIGS. **1** to **7** will be mainly described with reference to FIGS. **8**A, **8**B, and **9**.
(119) According to an embodiment of the disclosure, in the inner space (e.g., the space **801** of FIG.

8A) of the housing **200**, a rotary shaft **811** (or a rotary bar) may be disposed in the first direction of the electronic device **101** (the y direction). According to an embodiment of the disclosure, the rotary shaft **811** may be aligned to substantially pass through the center of the main body **210**. For example, the rotary shaft **811** may be disposed in parallel to the center of the first flat portion **220** and the center of the second flat portion **230**.

(120) According to an embodiment of the disclosure, the rotary shaft **811** may rotate based on a predetermined event. For example, a predetermined event may include various types of user inputs.

(121) According to an embodiment of the disclosure, in the inner space **801** of the housing **200**, a first PCB **821** (e.g., a main PCB) coupled to the rotary shaft **811** and rotating following the rotation of the rotary shaft **811** may be disposed. For example, when the rotary shaft **811** rotates clockwise, the first PCB **821** may rotate clockwise. For example, when the rotary shaft **811** rotates counterclockwise, the first PCB **821** may rotate counterclockwise.

(122) According to an embodiment of the disclosure, at least one battery **189** (e.g., the battery **189** of FIG. **1**) may be disposed in the inner space **801** of the housing **200** to face the first PCB **821**. For example, the battery **189** may be disposed on each of the opposite surfaces of the first PCB **821**. Referring to FIG. **8**B, the battery may include a first battery **189***a* disposed to face one surface of the first PCB **821** and a second battery **189***b* disposed to face the other surface of the first PCB **821**.

(123) According to an embodiment of the disclosure, the battery **189** may have a shape corresponding to that of the housing **200**. For example, the battery **189** may have a semi-cylindrical shape and may be disposed to face the first PCB **821**. According to an embodiment of the disclosure, the battery **189** may rotate together with the first PCB **821** following the rotation of the rotary shaft **811**.

(124) According to an embodiment of the disclosure, in the first state, at least a portion of the first display **241** may be accommodated in the inner space **801** of the housing **200**. According to an embodiment of the disclosure, the first display **241** is a rollable display and may be disposed in a state in which at least a portion of the first display **241** is rolled in the inner space **801** of the housing **200**. According to an embodiment of the disclosure, in the first state of the electronic device **101**, a portion of the first display **241** may be disposed in the inner space **801** of the housing **200** to surround the outer periphery of the first PCB **821**. According to an embodiment of the disclosure, the first display **241** may be pushed out from inside to outside of the housing **200** or pulled into inside from outside of the housing **200** in interlocking with the rotation of the rotary shaft **811**.

(125) In switching to the first state (e.g., the contracted state) and/or the second state (e.g., an extended state), the electronic device **101** according to various embodiments of the disclosure may be manually switched by a user or may be automatically switched by a drive mechanism (not illustrated) (e.g., a drive motor, a reduction gear module, and/or a gear assembly) disposed inside the housing **200**.

(126) According to an embodiment of the disclosure, an operation of the drive mechanism may be triggered based on a user input. According to an embodiment of the disclosure, the user input for triggering the operation of the drive mechanism may include a touch input, a force touch input, and/or a gesture input via the first display **241** or the second display **242**. In another embodiment of the disclosure, the user input for triggering the operation of the drive mechanism may include a voice input or an input by a physical button (e.g., a button **1611** in FIG. **16**) visually exposed outside through a portion of the main body **210**.

(127) According to an embodiment of the disclosure, a second PCB **822** may be disposed in the inner space **801** of the housing **200**. According to an embodiment of the disclosure, the second PCB **822** may be disposed between the first flat portion **220** and the first PCB **821** in the inner space **801** of the housing **200**. According to an embodiment of the disclosure, the second PCB **822** may be fixed by being coupled with a portion of the main body **210** in the inner space **801** of the housing **200**. According to an embodiment of the disclosure, the second PCB **822** may be fixed regardless

of the rotation of the rotary shaft **811** by being coupled to a portion of the main body **210**.

(128) Referring to FIG. **9**, the second PCB **822** according to an embodiment may be electrically connected to the conductive portion **610** of the first flat portion **220** via a first connecting member **911**. According to an embodiment of the disclosure, the conductive portion **610** may include the first conductive portion **611** used as a first antenna, the second conductive portion **612** used as a second antenna, and/or a third conductive portion **613** used as a third antenna. According to an embodiment of the disclosure, the second PCB **822** may be electrically connected to the first conductive portion **611**, the second conductive portion **612**, and/or the third conductive portion **613** via the first connecting member **911**. According to an embodiment of the disclosure, the first connecting member **911** may include a C-clip or a pogo pin.

(129) According to an embodiment of the disclosure, the first conductive portion **611**, the second conductive portion **612**, and/or the third conductive portion **613** arranged on the first flat portion **220** may be configured as a planner inverted F antenna (PIFA). According to an embodiment of the disclosure, the first conductive portion **611**, the second conductive portion **612**, and/or the third conductive portion **613** may be electrically connected to at least one feeding portion (not illustrated) disposed on the second PCB **822** and/or at least one ground via the first connecting member **911**.

(130) According to an embodiment of the disclosure, the first conductive portion **611** of the first flat portion **220** may be configured to operate as a first antenna by including a first feeding point **941-1** electrically connected to a feeding portion of the second PCB **822** via the first connecting member **911** and a first ground point **942-1** electrically connected to a ground of the second PCB **822** via the first connecting member **911**. According to an embodiment of the disclosure, the second conductive portion **612** of the first flat portion **220** may be configured to operate as a second antenna by including a second feeding point **941-2** electrically connected to the feeding portion of the second PCB **822** via the first connecting member **911** and a second ground point **942-2** electrically connected to the ground of the second PCB **822** via the first connecting member **911**.

(131) According to an embodiment of the disclosure, the third conductive portion **613** of the first flat portion **220** may be configured to operate as a third antenna by including a third feeding point **941-3** electrically connected to the feeding portion of the second PCB **822** via the first connecting member **911** and a switch portion **943** electrically connected to a switch of the second PCB **822** via the first connecting member **911**.

(132) According to an embodiment of the disclosure, a third PCB **823** may be disposed in the inner space **801** of the housing **200**. According to an embodiment of the disclosure, the third PCB **823** may be disposed between the second flat portion **230** and the first PCB **821** in the inner space **801** of the housing **200**. According to an embodiment of the disclosure, the third PCB **823** may be fixed by being coupled with a portion of the main body **210** in the inner space **801** of the housing **200**. According to an embodiment of the disclosure, the third PCB **823** may be fixed regardless of the rotation of the rotary shaft **811** by being coupled to a portion of the main body **210**. According to an embodiment of the disclosure, the third PCB **823** may be disposed to be substantially parallel to the second PCB **822**.

(133) According to an embodiment of the disclosure, the third PCB **823** may be electrically connected to the conductive portion **610** of the second flat portion **230** via a second connecting member (not illustrated). According to an embodiment of the disclosure, the conductive portion **610** may include the first conductive portion **611** used as a first antenna, the second conductive portion **612** used as a second antenna, and/or a third conductive portion **613** used as a third antenna. According to an embodiment of the disclosure, the third PCB **823** may be electrically connected to the first conductive portion **611**, the second conductive portion **612**, and/or the third conductive portion **613** via a second connecting member (not illustrated). According to an embodiment of the disclosure, the second connecting member (not illustrated) may include a C-clip or a pogo pin.

(134) According to an embodiment of the disclosure, the first conductive portion **611**, the second

conductive portion **612**, and/or the third conductive portion **613** arranged on the second flat portion **230** may be configured as a planner inverted F antenna (PIFA). According to an embodiment of the disclosure, the first conductive portion **611**, the second conductive portion **612**, and/or the third conductive portion **613** may be electrically connected to at least one feeding portion (not illustrated) disposed on the third PCB **823** and/or at least one ground via the second connecting member (not illustrated).
(135) According to an embodiment of the disclosure, a first rollable FPCB **831** may be disposed in the inner space **801** of the housing **200**. According to an embodiment of the disclosure, the first rollable FPCB **831** may electrically connect the first PCB **821** and the second PCB **822** to each other and may be spirally rolled around one end **811***a* of the rotary shaft **811**.
(136) According to an embodiment of the disclosure, a second rollable FPCB **832** may be disposed in the inner space **801** of the housing **200**. According to an embodiment of the disclosure, the second rollable FPCB **832** may electrically connect the first PCB **821** and the third PCB **823** to each other and may be spirally rolled around the other end **811***b* of the rotary shaft **811**.
(137) According to an embodiment of the disclosure, a first shield member **841** may be disposed between the first rollable FPCB **831** and the first PCB **821**. As another embodiment of the disclosure, a second shield member **842** may be disposed between the second rollable FPCB **832** and the first PCB **821**.
(138) According to an embodiment of the disclosure, the first rollable FPCB **831** may form a radiation pattern directed from the first shield member **841** toward the first flat portion **220** by including at least one wire (e.g., a plurality of wires **1241**, **1242**, and **1243** of FIG. **12**) configured to provide a wireless charging function.
(139) According to an embodiment of the disclosure, the second rollable FPCB **832** may form a radiation pattern directed from the second shield member **842** toward the second flat portion **230** by including at least one wire (e.g., the plurality of wires **1241**, **1242**, and **1243** of FIG. **12**) configured to provide a wireless charging function.
(140) FIG. **10** is a rear perspective view illustrating a contact structure of an antenna of an electronic device according to an embodiment of the disclosure.
(141) Referring to FIG. **10**, the electronic device **101** may include an embodiment that is at least partially similar to or different from the electronic device **101** illustrated in FIGS. **1** to **7**, **8**A, **8**B, and **9**. Hereinafter, features of the electronic device **101** that have not been described with reference to FIGS. **1** to **7**, **8**A, **8**B, and **9** or are changed from FIGS. **1** to **7**, **8**A, **8**B, and **9** will be mainly described with reference to FIG. **10**.
(142) According to an embodiment of the disclosure, the second PCB **822** may be disposed to face the first flat portion **220** and may have a circular first opening **1011** in the center thereof, thereby having a ring shape.
(143) According to an embodiment of the disclosure, the first rollable FPCB **831** may be spirally rolled and may have a diameter smaller than or equal to the diameter d**3** of the first opening **1011** in the second PCB **822**. For example, the maximum diameter d**4** of the rolled first rollable FPCB **831** may be smaller than or equal to the diameter d**3** of the first opening **1011** in the second PCB **822**. According to an embodiment of the disclosure, the first rollable FPCB **831** may be spirally rolled, and the maximum diameter d**4** of the outermost portion of the first rollable FPCB **831** may be smaller than or equal to the diameter d**3** of the first opening **1011** in the second PCB **822**.
(144) Although not illustrated, according to an embodiment of the disclosure, the third PCB **823** may be disposed to face the second flat portion **230** and may have a circular second opening (not illustrated) in the center thereof, thereby having a ring shape. According to an embodiment of the disclosure, the third PCB **823** may have substantially the same structure and shape as the second PCB **822**.
(145) Although not illustrated, the second rollable FPCB **832** according to an embodiment may be spirally rolled and may have a diameter smaller than or equal to the diameter of the second opening

in the third PCB **823**. For example, the maximum diameter d**4** of the rolled second rollable FPCB **832** may be smaller than or equal to the diameter of the second opening in the third PCB **823**. According to an embodiment of the disclosure, the second rollable FPCB **832** may be spirally rolled, and the maximum diameter d**4** of the outermost portion of the second rollable FPCB **832** may be smaller than or equal to the diameter of the second opening in the third PCB **823**. According to an embodiment of the disclosure, the second rollable FPCB **832** may have substantially the same structure and shape as the first rollable FPCB **831**.

(146) According to an embodiment of the disclosure, one end of the first rollable FPCB **831** may be disposed adjacent to one end **811***a* of the rotary shaft **811** and may be electrically connected to a portion of the first PCB **821**, and the other end of the first rollable FPCB **831** may be electrically connected to a portion of the second PCB **822**.

(147) Although not illustrated, according to an embodiment of the disclosure, one end of the second rollable FPCB **832** may be disposed adjacent to the other end **811***b* of the rotary shaft **811** and may be electrically connected to a portion of the first PCB **821**, and the other end of the second rollable FPCB **832** may be electrically connected to a portion of the third PCB **823**.

(148) According to an embodiment of the disclosure, the first rollable FPCB **831** may electrically connect a portion of the first PCB **821** and a portion of the second PCB **822** to each other in a connector-to-connector manner. According to an embodiment of the disclosure, the curvature and diameter of at least a portion of the first rollable FPCB **831** may be variable as the first PCB **821** and the rotary shaft **811** rotate. According to an embodiment of the disclosure, as the first PCB **821** and the rotary shaft **811** rotate clockwise (in a direction indicated by arrow **1001** in FIG. **10**), the surface of the first rollable FPCB **831** may increase in curvature and may decrease in diameter in at least a portion adjacent to the first PCB **821**. For example, as the first PCB **821** and the rotary shaft **811** rotate clockwise (in the direction indicated by arrow **1001** in FIG. **10**), the distance d**5** between a first portion of the first rollable PCB **831** disposed in a k.sup.th turn and a second portion of the first rollable FPCB **831** disposed in a (k+1).sup.th turn of the first rollable FPCB **831** may decrease. For example, the number of turns of the first rollable FPCB **831** may increase as the first PCB **821** and the rotary shaft **811** rotate clockwise (in the direction indicated by arrow **1001** in FIG. **10**).

(149) According to an embodiment of the disclosure, as the first PCB **821** and the rotary shaft **811** rotate counterclockwise (in a direction indicated by arrow **1002** in FIG. **10**), the surface of the first rollable FPCB **831** may decrease in curvature and may increase in diameter in at least a portion adjacent to the first PCB **821**. For example, as the first PCB **821** and the rotary shaft **811** rotate counterclockwise (in the direction indicated by arrow **1001** in FIG. **10**), the distance d**5** between the first portion of the first rollable PCB **831** disposed in the k.sup.th turn and the second portion of the first rollable FPCB **831** disposed in the (k+1).sup.th turn of the first rollable FPCB **831** may increase.

(150) According to various embodiments of the disclosure, the curvature of the surface of the first rollable FPCB **831** may be variable depending on whether the electronic device **101** is in the first state (e.g., the slide-in state) or in the second state (e.g., the slide-out state).

(151) According to an embodiment of the disclosure, when the electronic device **101** is switched from the first state to the second state, the first PCB **821** and the rotary shaft **811** may rotate counterclockwise (in the direction indicated by arrow **1002** in FIG. **10**). In this case, the curvature of the surface of the first rollable FPCB **831** may decrease in interlocking with the switching of the electronic device **101** from the first state to the second state.

(152) According to an embodiment of the disclosure, when the electronic device **101** is switched from the second state to the first state, the first PCB **821** and the rotary shaft **811** may rotate clockwise (in the direction indicated by arrow **1001** in FIG. **10**). In this case, the curvature of the surface of the first rollable FPCB **831** may increase in interlocking with the switching of the electronic device **101** from the first state to the second state.

(153) According to another embodiment of the disclosure, when the electronic device **101** is

switched from the first state to the second state, the first PCB **821** and the rotary shaft **811** may rotate clockwise (in the direction indicated by arrow **1001** in FIG. **10**). In this case, the curvature of the surface of the first rollable FPCB **831** may increase in interlocking with the switching of the electronic device **101** from the first state to the second state.
(154) According to another embodiment of the disclosure, when the electronic device **101** is switched from the second state to the first state, the first PCB **821** and the rotary shaft **811** may rotate counterclockwise (in the direction indicated by arrow **1002** in FIG. **10**). In this case, the curvature of the surface of the first rollable FPCB **831** may decrease in interlocking with the switching of the electronic device **101** from the first state to the second state.
(155) Although not illustrated, the second rollable FPCB **832** according to an embodiment may electrically connect a portion of the first PCB **821** and a portion of the third PCB **823** to each other in a connector-to-connector manner. According to an embodiment of the disclosure, the curvature and diameter of at least a portion of the second rollable FPCB **832** may be variable as the first PCB **821** and the rotary shaft **811** rotate. According to an embodiment of the disclosure, as the first PCB **821** and the rotary shaft **811** rotate clockwise, the surface of the second rollable FPCB **832** may increase in curvature and may decrease in diameter in at least a portion adjacent to the first PCB **821**. For example, as the first PCB **821** and the rotary shaft **811** rotate clockwise, the distance between a first portion of the second rollable PCB **832** disposed in the k.sup.th turn and a second portion of the second rollable FPCB **832** disposed in the (k+1).sup.th turn of the second rollable FPCB **832** may decrease. According to an embodiment of the disclosure, as the first PCB **821** and the rotary shaft **811** rotate counterclockwise, the surface of the second rollable FPCB **832** may decrease in curvature and may increase in diameter in at least a portion adjacent to the first PCB **821**. For example, as the first PCB **821** and the rotary shaft **811** rotate clockwise, the distance between the first portion of the second rollable PCB **832** disposed in the k.sup.th turn and the second portion of the second rollable FPCB **832** disposed in the (k+1).sup.th turn of the second rollable FPCB **832** may increase.
(156) Although not illustrated, according to an embodiment of the disclosure, when the electronic device **101** is switched from the first state to the second state, the first PCB **821** and the rotary shaft **811** may rotate counterclockwise (in the direction indicated by arrow **1002** in FIG. **10**). In this case, the curvature of the surface of the second rollable FPCB **832** may decrease in interlocking with the switching of the electronic device **101** from the first state to the second state.
(157) Although not illustrated, according to an embodiment of the disclosure, when the electronic device **101** is switched from the second state to the first state, the first PCB **821** and the rotary shaft **811** may rotate clockwise (in the direction indicated by arrow **1001** in FIG. **10**). In this case, the curvature of the surface of the second rollable FPCB **832** may increase in interlocking with the switching of the electronic device **101** from the first state to the second state.
(158) Although not illustrated, according to another embodiment of the disclosure, when the electronic device **101** is switched from the first state to the second state, the first PCB **821** and the rotary shaft **811** may rotate clockwise (in the direction indicated by arrow **1001** in FIG. **10**). In this case, the curvature of the surface of the second rollable FPCB **832** may increase in interlocking with the switching of the electronic device **101** from the first state to the second state.
(159) Although not illustrated, according to another embodiment of the disclosure, when the electronic device **101** is switched from the second state to the first state, the first PCB **821** and the rotary shaft **811** may rotate counterclockwise (in the direction indicated by arrow **1002** in FIG. **10**). In this case, the curvature of the surface of the second rollable FPCB **832** may decrease in interlocking with the switching of the electronic device **101** from the first state to the second state.
(160) FIG. **11**A is a perspective view of an electronic device illustrating guide wheels configured to guide a movement of a rollable FPCB according to an embodiment of the disclosure.
(161) FIG. **11**B is a cross-sectional view of an electronic device illustrating a cross section of a guide wheel according to an embodiment of the disclosure.

(162) Referring to FIGS. **11**A and **11**B, the electronic device **101** may include an embodiment that is at least partially similar to or different from the electronic device **101** illustrated in FIGS. **1** to **7**, **8**A, **8**B, **9**, and **10**. Hereinafter, features of the electronic device **101** that have not been described with reference to FIGS. **1** to **7**, **8**A, **8**B, **9**, and **10** or are changed from FIGS. **1** to **7**, **8**A, **8**B, **9**, and **10** will be mainly described with reference to FIGS. **11**A and **11**B.
(163) Referring to FIGS. **11**A and **11**B, an electronic device (e.g., the electronic device **101** of FIG. **1**) according to an embodiment may include one or more guide wheels **1110** configured to guide the movement of a rollable FPCB (e.g., the first rollable FPCB **831** or the second roller FPCB **832** of FIG. **5**).
(164) According to an embodiment of the disclosure, the guide wheel **1110** may be disposed in a first opening **1011** (e.g., the first opening **1011** in FIG. **10**) in the center of the second PCB **822** and may rotate along the surface of the first rollable FPCB **831**. According to an embodiment of the disclosure, the guide wheel **1110** may be configured to rotate along the surface of the first rollable FPCB **831** and/or the second rollable FPCB **832**, thereby reducing damage to the first rollable FPCB **831** and/or the second rollable FPCB **831**. For example, as described above, the curvature and diameter of at least a portion of the first rollable FPCB **831** and/or the second rollable FPCB **832** may be variable as the rotary shaft **811** rotates. The guide wheels **1110** according to an embodiment may reduce an impact applied to the first rollable FPCB **831** and/or the second rollable FPCB **832** when the first rollable FPCB **831** and/or the second rollable FPCB **832** are bent or rolled as described above, thereby reducing the damage to the first rollable FPCB **831** and/or the second rollable FPCB **832**.
(165) Although not illustrated, the guide wheels **1110** may be disposed in the second opening (not illustrated) in the center of the third PCB **823** and may rotate along the surface of the second rollable FPCB **832**.
(166) Referring to FIG. **11**A, the guide wheels **1110** according to an embodiment may be disposed to be movable along rail members **1120**, respectively. According to an embodiment of the disclosure, the rail members **1120** may be disposed respectively from outer portions of the first openings **1011** adjacent to the second PCB **822** to be directed toward the central portions adjacent to the rotary shaft **811** of the electronic device **101**. Accordingly, the guide wheels **1110** may reciprocate respectively between the outer portions of the first openings **1011** adjacent to the second PCB **822** and the central portions adjacent to the rotary shaft **811** of the electronic device **101**. According to various embodiments of the disclosure, the electronic device **101** may include one or more rail members **1120** corresponding to the number of guide wheels **1110**, and the one or more rail members **1120** may be arranged to be directed toward the rotary shaft **811** of the electronic device **101**. According to various embodiments of the disclosure, the number of guide wheels **1110** is not limited to that illustrated in the drawings.
(167) Referring to FIG. **11**B, each guide wheel **1110** according to an embodiment may include a wheel member **1110** configured to come into contact with the surface of the first rollable FPCB **831** and/or the second rollable FPCB **832** and a guide shaft **1112** penetrating the wheel member **1110**. According to an embodiment of the disclosure, the guide shaft **1112** of the guide wheel **1110** may move along a rail member **1120**.
(168) According to an embodiment of the disclosure, the rail member **1120** may be configured to guide the movement of opposite ends of the guide shaft **1112** of the guide wheel **1110**. The rail member **1120** according to an embodiment may include a first rail member **1121** provided on the first flat portion **220** and a second rail member **1122** provided on the first shield member **841**.
(169) According to an embodiment of the disclosure, the first rail member **1121** may be provided on the first flat portion **220** and may be configured to guide one end **1112***a* of the guide shaft **1112** of the guide wheel **1110** to reciprocate between an outer portion of the first opening **1011** and a central portion adjacent to the rotary shaft (e.g., the rotary shaft **811** of FIG. **11**B) of the electronic device **101**.

(170) According to an embodiment of the disclosure, the second rail member **1122** may be provided on the first shield member **841** and may be configured to guide the other end **1112***b* of the guide shaft **1112** of the guide wheel **1110** to reciprocate between the outer portion of the first opening **1011** and the central portion adjacent to the rotary shaft **811** of the electronic device **101**.
(171) Although not illustrated, the first rail member **1121** and the second rail member **1122** according to an embodiment may be similarly provided on the second flat portion **230**. For example, although not illustrated, the first rail member **1121** according to an embodiment may be provided on the second flat portion **230** and may be configured to guide the one end **1112***a* of the guide shaft **1112** of the guide wheel **1110** to reciprocate between an outer portion of a second opening (not illustrated) and a central portion adjacent to the rotary shaft (e.g., the rotary shaft **811** of FIG. **11**B) of the electronic device **101**. For example, although not illustrated, the second rail member **1122** according to an embodiment may be provided on the second shield member **842** and may be configured to guide the other end **1112***b* of the guide shaft **1112** of the guide wheel **1110** to reciprocate between the outer portion of the first opening **1011** and the central portion adjacent to the rotary shaft **811** of the electronic device **101**.
(172) FIG. **12** is a plan view schematically illustrating a structure of a rollable FPCB according to an embodiment of the disclosure.
(173) The rollable FPCB illustrated in FIG. **12** may include an embodiment that is at least partially similar to or different from the first rollable FPCB or the second rollable FPCB illustrated in FIG. **8**A. Hereinafter, features of the rollable FPCB that have not been described with reference to FIGS. **1** to **7**, **8**A, **8**B, **9**, **10**, **11**A and **11**B or are changed from FIGS. **1** to **7**, **8**A, **8**B, **9**, **10**, **11**A and **11**B will be mainly described with reference to FIG. **12**.
(174) Referring to FIG. **12**, a rollable FPCB (e.g., the first rollable FPCB **831** and/or the second rollable FPCB **832** of FIG. **8**A) according to an embodiment may include one or more connector areas, one or more nonelastic areas, and/or one or more elastic areas.
(175) According to an embodiment of the disclosure, each connector area **1210** is an area including a connector (not illustrated) and/or a pad (not illustrated) and may include a first connector area **1211** coupled to a first PCB (e.g., the first PCB in FIG. **8**A) and/or a second connector area **1212** coupled to a second PCB (e.g., the second PCB **822** in FIG. **8**A or the third PCB **823** in FIG. **8**A).
(176) According to an embodiment of the disclosure, each nonelastic area **1220** may include a rigid PCB made of a rigid material (or a non-flexible material). According to an embodiment of the disclosure, at least one nonelastic area **1220** may be disposed between the first connector area **1211** and the second connector area **1212**. For example, the nonelastic region **1220** may be disposed between the first connector area **1211** and the second connector area **1212** to correspond to a portion in which the rollable FPCB (e.g., the first rollable FPCB **831** and/or the second roller FPCB of FIG. **8**A) is bent.
(177) According to an embodiment of the disclosure, an elastic area **1230** may be disposed between the connector area **1210** and the nonelastic area **1220** and may include the plurality of wires **1241**, **1242**, and **1243**. According to an embodiment of the disclosure, the plurality of wires **1241**, **1242**, and **1243** disposed in the elastic area **1230** may include at least one signal wire **1241** configured to transmit an RF signal or a control signal, a wireless charging wire **1242** configured to provide a wireless charging function, and/or at least one ground wire **1243** electrically connected to a ground. According to an embodiment of the disclosure, the ground wire **1243** may be disposed between the signal wire **1241** and the wireless charging wire **1242**. For example, the wireless charging wire **1242** and the at least one signal wire **1241** are spaced apart from each other with the ground wire **1243** interposed therebetween, thereby reducing signal interference or noise.
(178) According to an embodiment of the disclosure, the wireless charging wire **1242** may be disposed at the outermost portion among the plurality of wires **1241**, **1242**, and **1243**. For example, among the plurality of wires **1241**, **1242**, and **1243**, the wireless charging wire **1242** may be disposed closest to the first flat portion **220** or the second flat portion **230**. When the wireless

charging wire **1242** and a first shield member (e.g., the first shield member **841** of FIG. **8**A) (or the second shield member (e.g., the second shield member **842** of FIG. **8**A)) is disposed close to the first flat portion **220** or the second flat portion **230**, the electronic device **101** according to an embodiment may provide the wireless charging function via the first flat portion **220** or the second flat portion **230**. For example, the battery (e.g., the battery **189** in FIG. **1**) of the electronic device **101** can be charged wirelessly when the first flat portion **220** or the second flat portion **230** is placed on a wireless charging pad (or a wireless charging cradle) (not illustrated) configured to feed power wirelessly.

(179) FIG. **13** is a view illustrating a structure of an antenna of an electronic device according to an embodiment of the disclosure.

(180) The electronic device **101** illustrated in FIG. **13** may include an embodiment that is at least partially similar to or different from the electronic device **101** illustrated in FIGS. **1** to **7**, **8**A, **8**B, **9**, **10**, **11**A, **11**B, and **12**. Hereinafter, features of the electronic device **101** that have not been described with reference to FIGS. **1** to **7**, **8**A, **8**B, **9**, **10**, **11**A, **11**B, and **12** or are changed from FIGS. **1** to **7**, **8**A, **8**B, **9**, **10**, **11**A, **11**B, and **12** will be mainly described with reference to FIG. **13**.

(181) Referring to FIG. **13**, the first flat portion (e.g., the first flat portion **220** of FIG. **8**A) or the second flat portion (e.g., the second flat portion **230** of FIG. **8**A) according to an embodiment may include the first conductive portion **611**, the second conductive portion **612**, and/or a third conductive portion **613**, which may be split by the non-conductive portion **620**.

(182) According to an embodiment of the disclosure, the first conductive portion **611** may operate as a first antenna radiator by including a first feeding point **941-1** electrically connected to a feeding portion of a second PCB (e.g., the second PCB **822** of FIG. **8**A) (or the third PCB **823** of FIG. **8**A) and a first ground point **942-1** electrically connected to a ground (not illustrated) of the second PCB **822**. According to an embodiment of the disclosure, the second conductive portion **612** may operate as a second antenna radiator by including a second feeding point **941-2** electrically connected to the feeding portion of the second PCB **822** (or the third PCB **823** of FIG. **8**A) and a second ground point **942-2** electrically connected to the ground of the second PCB **822**. According to an embodiment of the disclosure, the third conductive portion **613** may operate as a third antenna by including a third feeding point **941-3** electrically connected to the feeding portion of the second PCB **822** (or the third PCB **823** in FIG. **8**A) and a switch connecting point **943** electrically connected to a switch of the second PCB **822** (or the third PCB **823** in FIG. **8**A).

(183) FIG. **14** is a view illustrating a structure of an antenna of an electronic device according to an embodiment of the disclosure.

(184) The electronic device **101** illustrated in FIG. **14** may include an embodiment that is at least partially similar to or different from the electronic device **101** illustrated in FIGS. **1** to **7**, **8**A, **8**B, **9**, **10**, **11**A, **11**B, **12**, and **13**. Hereinafter, features of the electronic device **101** that have not been described with reference to FIGS. **1** to **7**, **8**A, **8**B, **9**, **10**, **11**A, **11**B, **12**, and **13** or are changed from FIGS. **1** to **7**, **8**A, **8**B, **9**, **10**, **11**A, **11**B, **12**, and **13** will be mainly described with reference to FIG. **14**.

(185) Referring to FIG. **14**, an electronic device **101** according to another embodiment may include extensions **611***b*, **612***b*, and **613***b* obtained by extending lengths of one or more of the first conductive portion **611**, the second conductive portion **612**, and/or the third conductive portion **613** toward the rotary shaft **811**. For example, when conductive portions serving as antennas are disposed only in the outer edge area of the first flat portion (e.g., the first flat portion **220** in FIG. **8**A) or the second flat portion (e.g., the second flat portion **230**), the length of antennas may be insufficient in the electronic device **101**. In the electronic device **101** according to another embodiment of the disclosure, by providing the extensions **611***b*, **612***b*, and **613***b* having lengths extending toward the rotary shaft **811** for one or more of the first conductive portion **611**, the second conductive portion **612**, and/or the third conductive portion **613**, it is possible to increase the length of antennas.

(186) In the electronic device **101** according to another embodiment of the disclosure, since the first conductive portion **611**, the second conductive portion **612**, and/or the third conductive portion **613** include extensions **611***b*, **612***b*, and **613***b*, it is possible to provide antennas having one or more slit structures **1411**, **1412**, and **1413**. In an embodiment of the disclosure, the shapes of the extensions **611***b*, **612***b*, and **613***b* of the first conductive portion **611**, the second conductive portion **612**, and/or the third conductive portion **613** may be determined based on the operating frequencies of antennas in which the first conductive portion **611**, the second conductive portion **612**, and/or the third conductive portion **613** include the extension portions **611***b*, **612***b*, and **613***b*.
(187) According to another embodiment of the disclosure, the first conductive portion **611** may include a first outer portion **611***a* disposed in the outer edge area of the first flat portion (e.g., the first flat portion **220** of FIG. **8**A) or the second flat portion (e.g., the second flat portion **230** of FIG. **8**A) and a first extension **611***b* extending from the first outer portion **611***a* toward the rotary shaft **811**, and a first feeding point **941-1** electrically connected to a feeding portion of the second PCB **822** (or the third PCB **823** in FIG. **8**A) may be provided in the first extension **611***b*. For example, a first slit structure **1411** made of a non-conductive material may be provided between the first outer portion **611***a* and the first extension **611***b*.
(188) According to another embodiment of the disclosure, the second conductive portion **612** may include a second outer portion **612***a* disposed in the outer edge area of the first flat portion (e.g., the first flat portion **220** of FIG. **8**A) or the second flat portion (e.g., the second flat portion **230** of FIG. **8**A) and a second extension **612***b* extending from the second outer portion **612***a* toward the rotary shaft **811**, and a second feeding point **941-2** electrically connected to a feeding portion of the second PCB **822** (or the third PCB **823** in FIG. **8**A) may be provided in the second extension **612***b*. For example, a second slit structure **1412** made of a non-conductive material may be provided between the second outer portion **612***a* and the second extension **612***b*.
(189) According to another embodiment of the disclosure, the third conductive portion **613** may include a third outer portion **613***a* disposed in the outer edge area of the first flat portion (e.g., the first flat portion **220** of FIG. **8**A) or the second flat portion (e.g., the second flat portion **230** in FIG. **8**A) and a third extension **613***b* extending from the third outer portion **613***a* toward the rotary shaft **811**, and a third feeding point **941-3** electrically connected to a feeding portion of the second PCB **822** (or the third PCB **823** in FIG. **8**A) and a ground point **942** electrically connected to a ground may be provided in the third extension **613***b*. For example, a third slit structure **1413** made of a non-conductive material may be provided between the third outer portion **613***a* and the third extension **613***b*.
(190) FIG. **15** is a view illustrating a structure of an antenna of an electronic device according to an embodiment of the disclosure.
(191) The electronic device **101** illustrated in FIG. **15** may include an embodiment that is at least partially similar to or different from the electronic device **101** illustrated in FIGS. **1** to **7**, **8**A, **8**B, **9**, **10**, **11**A, **11**B, **12**, **13**, and **14**. Hereinafter, features of the electronic device **101** that have not been described with reference to FIGS. **1** to **7**, **8**A, **8**B, **9**, **10**, **11**A, **11**B, **12**, **13**, and **14** or are changed from FIGS. **1** to **7**, **8**A, **8**B, **9**, **10**, **11**A, **11**B, **12**, **13**, and **14** will be mainly described with reference to FIG. **15**.
(192) Referring to FIG. **15**, the first flat portion (e.g., the first flat portion **220** of FIG. **8**A) or the second flat portion (e.g., the second flat portion **230** of FIG. **8**A) according to another embodiment may include a first conductive portion **1511**, a second conductive portion **1512**, a third conductive portion **1513**, a fourth conductive portion **1514**, and/or a fifth conductive portion **1515**, which may be split by the non-conductive portion **620**.
(193) According to another embodiment of the disclosure, by providing one or more extensions **1511***b*, **1512***b*, **1513***b*, **1514***b*, and **1515***b* having lengths extending toward the rotary shaft **811** for one or more of the first conductive portion **1511**, the second conductive portion **1512**, the third conductive portion **1513**, the fourth conductive portion **1514**, and/or the fifth conductive portion

1515, it is possible to increase the lengths of antennas or to adjust a length of a peripheral antenna.
(194) According to another embodiment of the disclosure, the first conductive portion **1511** may include a first outer portion **1511***a* disposed in the outer edge area of the first flat portion (e.g., the first flat portion **220** of FIG. **8**A) or the second flat portion (e.g., the second flat portion **230** of FIG. **8**A) and a first extension **1511***b* extending from the first outer portion **1511***a* toward the rotary shaft **811**. By providing a first feeding point **941-1** electrically connected to the feeding portion of the second PCB **822** (or the third PCB **823** in FIG. **8**A) in the first outer portion **1511***a*, the first conductive portion **1511** may operate as a first antenna.
(195) According to another embodiment of the disclosure, the second conductive portion **1512** may include a second outer portion **1512***a* disposed in the outer edge area of the first flat portion (e.g., the first flat portion **220** of FIG. **8**A) or the second flat portion (e.g., the second flat portion **230** of FIG. **8**A) and a second extension **1512***b* extending from the second outer portion **1512***a* toward the rotary shaft **811**. By providing a second feeding point **941-2** electrically connected to the feeding portion of the second PCB **822** (or the third PCB **823** in FIG. **8**A) in the second outer portion **1512***a*, the second conductive portion **1512** may operate as a second antenna.
(196) According to another embodiment of the disclosure, the third conductive portion **1513** may include a third outer portion **1513***a* disposed in the outer edge area of the first flat portion (e.g., the first flat portion **220** of FIG. **8**A) or the second flat portion (e.g., the second flat portion **230** of FIG. **8**A) and a third extension **1513***b* extending from the third outer portion **1513***a* toward the rotary shaft **811**. By providing a third feeding point **941-3** electrically connected to the feeding portion of the second PCB **822** (or the third PCB **823** in FIG. **8**A) in the third outer portion **1513***a*, the third conductive portion **1513** may operate as a third antenna.
(197) According to another embodiment of the disclosure, the fourth conductive portion **1514** may include a fourth outer portion **1514***a* disposed in the outer edge area of the first flat portion (e.g., the first flat portion **220** of FIG. **8**A) or the second flat portion (e.g., the second flat portion **230** of FIG. **8**A) and a fourth extension **1514***b* extending from the fourth outer portion **1514***a* toward the rotary shaft **811**. By providing a fourth feeding point **941-4** electrically connected to the feeding portion of the second PCB **822** (or the third PCB **823** in FIG. **8**A) and a ground point **942** electrically connected to a ground in the fourth outer portion **1514***a*, the fourth conductive portion **1514** may operate as a fourth antenna.
(198) According to another embodiment of the disclosure, the fifth conductive portion **1515** may include a fifth outer portion **1515***a* disposed in the outer edge area of the first flat portion (e.g., the first flat portion **220** of FIG. **8**A) or the second flat portion (e.g., the second flat portion **230** of FIG. **8**A) and a fifth extension **1515***b* extending from the fifth outer portion **1515***a* toward the rotary shaft **811**. By providing a switch portion **943** electrically connected to a switch of the second PCB **822** (or the third PCB **823** in FIG. **8**A) in the fifth extension **1515***b*, the fifth conductive portion **1515** may provide an antenna tuning structure configured to adjust the length of a peripheral antenna.
(199) FIG. **16** is a plan view illustrating physical buttons of an electronic device according to an embodiment of the disclosure.
(200) The electronic device **101** illustrated in FIG. **16** may include an embodiment that is at least partially similar to or different from the electronic device **101** illustrated in FIGS. **1** to **7**, **8**A, **8**B, **9**, **10**, **11**A, **11**B, **12**, **13**, **14**, and **15**. Hereinafter, features of the electronic device **101** that have not been described with reference to FIGS. **1** to **7**, **8**A, **8**B, **9**, **10**, **11**A, **11**B, **12**, **13**, **14**, and **15** or are changed from FIGS. **1** **7**, **8**A, **8**B, **9**, **10**, **11**A, **11**B, **12**, **13**, **14**, and to **15** will be mainly described with reference to FIG. **16**.
(201) Referring to FIG. **16**, the electronic device **101** according to various embodiments may include one or more physical buttons **1611** visibly exposed through at least a portion of the housing **200**, for example, at least a portion of the main body **210**.
(202) According to various embodiments of the disclosure, the one or more physical buttons **1611**

may include at least one of a volume control button configured to adjust the volume of a speaker, a power button, a function button configured to control an artificial intelligence application, or a user custom button configured to provide a user custom function. In some embodiments of the disclosure, the electronic device **101** may be switched from the first state to the second state or from the second state to the first state by triggering the operation of a drive mechanism based on a user input made through the one or more physical buttons **1611**.
(203) Reference numeral **241**, which has not been described in FIG. **16**, may denote a first display (e.g., the first display **241** of FIG. **3**).
(204) Reference numeral **242**, which has not been described in FIG. **16**, may denote a second display (e.g., the second display **242** of FIG. **3**).
(205) While the disclosure has been shown and described with reference to various embodiments thereof, it will be understood by those skilled in the art that various changes in form and details may be made therein without departing from the spirit and scope of the disclosure as defined by the appended claims and their equivalents.

Claims

1. An electronic device comprising: a housing comprising: a first flat portion disposed at one end of the housing, and a second flat portion disposed at another end of the housing and disposed in parallel to the first flat portion, wherein each of the first flat portion and the second flat portion comprises at least one conductive portion and at least one non-conductive portion; a rotary shaft disposed in a first direction in an inner space of the housing; a first printed circuit board (PCB) coupled to the rotary shaft and configured to rotate following rotation of the rotary shaft; a rollable display arranged to be rolled in the inner space of the housing and pushed out from inside to outside of the housing in interlocking with the rotation of the rotary shaft in a second direction perpendicular to the first direction; a first electrical component disposed between the first flat portion and one end of the rotary shaft in the inner space of the housing and fixed regardless of the rotation of the rotary shaft, wherein the first electrical component is operably connected to the at least one conductive portion of the first flat portion via a first connecting member; and a first rollable flexible PCB (FPCB) electrically connecting the first PCB and the first electrical component to each other, wherein the first rollable FPCB is spirally rolled around the one end of the rotary shaft.
2. The electronic device of claim 1, wherein one end of the first rollable FPCB is disposed adjacent to the one end of the rotary shaft and is electrically connected to a portion of the first PCB, and wherein another end of the first rollable FPCB is electrically connected to a portion of the first electrical component.
3. The electronic device of claim 1, wherein the first electrical component comprises a circular first opening in a center thereof, and wherein the first rollable FPCB has a diameter smaller than or equal to a diameter of the circular first opening.
4. The electronic device of claim 1, wherein a first shield member is disposed between the first rollable FPCB and the first PCB, and wherein the first rollable FPCB is configured to form a radiation pattern in a direction from the first shield member toward the first flat portion by comprising at least one wire configured to provide a wireless charging function.
5. The electronic device of claim 1, wherein the housing is configured in a hollow shape which comprises one of a cylindrical shape or a prismatic shape, and wherein the rollable display is accommodated in the inner space of the housing in a rolled state.
6. The electronic device of claim 1, further comprising: a second electrical component disposed between the second flat portion and another end of the rotary shaft in the inner space of the housing and fixed regardless of the rotation of the rotary shaft, wherein the second electrical component is operably connected to the at least one conductive portion of the second flat portion via a second

connecting member; and a second rollable FPCB electrically connecting the first PCB and the second electrical component to each other, wherein the second rollable FPCB is spirally rolled around the other end of the rotary shaft.

7. The electronic device of claim 6, wherein one end of the second rollable FPCB is disposed adjacent to the other end of the rotary shaft and is electrically connected to a portion of the first PCB, and wherein the other end of the second rollable FPCB is electrically connected to a portion of the second electrical component.

8. The electronic device of claim 6, wherein the first PCB comprises a circular second opening in a center thereof, and wherein the second rollable FPCB has a diameter smaller than or equal to a diameter of the circular second opening.

9. The electronic device of claim 6, wherein a second shield member is disposed between the second rollable FPCB and the first PCB.

10. The electronic device of claim 9, wherein the second rollable FPCB comprises at least one wire configured to provide a wireless charging function, thereby forming a radiation pattern in a direction from the second shield member toward the second flat portion.

11. The electronic device of claim 1, further comprising a plurality of guide wheels spirally disposed along a surface of the first rollable PCB at intervals, wherein each of the plurality of guide wheels comprises: a wheel member that comes into contact with the surface of the first rollable PCB; and a guide shaft penetrating the wheel member.

12. The electronic device of claim 11, further comprising a plurality of rail members configured to guide movement of the plurality of guide wheels, wherein the plurality of rail members are provided to be directed toward a center of the rotary shaft to enable the plurality of guide wheels to reciprocate toward the center of the rotary shaft.

13. The electronic device of claim 12, wherein the plurality of rail members comprises: at least one first rail member provided on the first flat portion; and a second rail member provided on a first shield member disposed between the first rollable FPCB and the first PCB.

14. The electronic device of claim 1, wherein the at least one conductive portion comprises: a first conductive portion; a second conductive portion; and a third conductive portion disposed at intervals along an outer edge of the first flat portion to be split by the non-conductive portion.

15. The electronic device of claim 14, wherein the first conductive portion is configured to operate as a first antenna by comprising a first feeding point electrically connected to a feeding portion of the first electrical component and a first ground point electrically connected to a ground, wherein the second conductive portion is configured to operate as a second antenna by comprising a second feeding point electrically connected to the feeding portion of the first electrical component and a second ground point electrically connected to the ground, and wherein the third conductive portion is configured to operate as a third antenna by comprising a third feeding point electrically connected to the feeding portion of the first electrical component and a switch portion electrically connected to a switch of the first electrical component.